

PROJECT DOCUMENTATION

FLEXGENIE

AI-Powered Dynamic Workout & Nutrition Plan Generator

BCA Team Academic Project (2026–2027)

PROJECT PURPOSE

FlexGenie is designed to leverage advanced generative AI models to dynamically generate hyper-personalized fitness routines, progressive resistance training schedules, and custom macronutrient meal plans tailored to individual body metrics, dietary restrictions, and specific fitness goals.

Degree Program: Bachelor of Computer Applications (BCA)

Academic Year: 2026–2027

Institution: Department of Computer Science & Applications

Certificate of Approval

This is to certify that the project entitled "**FlexGenie: AI-Powered Dynamic Workout & Nutrition Plan Generator**" is a bonafide record of team project work carried out as part of the academic requirements for the Bachelor of Computer Applications degree.

The project successfully implements modern software development methodologies, responsive UI design, backend integration, and structured AI response parsing.

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1. INTRODUCTION & OBJECTIVES

Traditional fitness applications offer static, pre-packaged workout routines and generic calorie calculators that fail to adapt to user feedback, injury limitations, or fluctuating lifestyle constraints. **FlexGenie** solves this gap by integrating generative AI models to construct intelligent, context-aware training regimens.

Key Objectives

- Develop an intuitive web and mobile-responsive interface for user onboarding and metrics collection.
- Implement secure user authentication and profile management for tracking body weight, target goals, and equipment availability.
- Design prompt-engineering pipelines to query large language models with structured JSON constraints for clean parsing.
- Generate detailed weekly workout split schedules with exercise descriptions, sets, reps, and rest intervals.
- Provide customized nutrition breakdowns including daily caloric targets and macro distribution (***Protein***, ***Carbs***, ***Fats***).

2. SYSTEM ARCHITECTURE & TECH STACK

FlexGenie utilizes a modern, modular 3-tier architecture separating the client presentation layer, backend application server, and the external AI generation pipeline.

Technology Stack

- **Frontend:** React.js, Tailwind CSS for responsive styling, Lucide Icons.
- **Backend:** Node.js with Express framework, RESTful API endpoints.
- **AI Integration:** Google Gemini API for natural language reasoning and structured workout generation.
- **Database:** MongoDB for secure user credential storage and history logging.

3. CORE FUNCTIONAL MODULES

The application is structured into four primary interactive modules:

- **Onboarding Questionnaire:** Collects user age, gender, height, current weight, target weight, activity level, dietary preferences (Vegan, Keto, Omnivore), and available equipment (Gym, Home Dumbbells, Bodyweight).
- **AI Prompt Synthesizer:** Converts user parameters into optimized system prompts with strict JSON schema constraints.
- **Workout Dashboard:** Displays daily exercises with interactive timers, video demonstration links, and weight/rep loggers.
- **Progress Tracker:** Visualizes weight change trends over time and logs completed workout milestones.

4. CONCLUSION & FUTURE ENHANCEMENTS

FlexGenie successfully demonstrates the practical implementation of AI-driven personalization in health and wellness domains. Future versions will incorporate computer vision modules for real-time form correction using device cameras and wearable device synchronization via smartwatch APIs.